



# DW\_fp\_log2

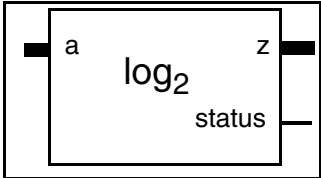
## Floating-Point Base-2 Logarithm

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

### Features and Benefits

- The precision is controlled by parameters, and covers formats in the IEEE Standard 754
- Exponents can range from 3 to 31 bits
- Fractional part of the floating point number can range from 2 to 60 bits
- A parameter controls the use of denormal values

### Revision History



### Description

This component computes the base-2 logarithm of a floating-point input *a*, delivering an output *z* that is also a floating-point value.

The parameter *ieee\_compliance* controls the use of denormals and NaNs, as done for other FP operators in this library. When *ieee\_compliance* = 0, the operator takes NaN values as infinities, and denormals as zeros. When *ieee\_compliance* = 1, the component accepts and generates denormalized values, handles NaN inputs, and delivers NaN outputs when appropriate. Independently of the value of this parameter, the floating-point format can be adjusted to match one of the formats defined in the IEEE Standard 754.

The output *status* is an 8-bit value that carries the status flags for the FP operation, as described in the [Datapath Floating-Point Overview](#).

Table 1-1 Pin Description

| Pin Name | Width                                           | Direction | Function                                                                                                                                                                                                                                      |
|----------|-------------------------------------------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| a        | ( <i>sig_width</i> + <i>exp_width</i> + 1) bits | Input     | Input data.                                                                                                                                                                                                                                   |
| z        | ( <i>sig_width</i> + <i>exp_width</i> + 1) bits | Output    | log <sub>2</sub> ( <i>a</i> )                                                                                                                                                                                                                 |
| status   | 8-bits                                          | Output    | <ul style="list-style-type: none"><li>■ Status flags for the result; for details, see <a href="#">STATUS Flags</a> in the <i>Datapath Floating-Point Overview</i></li><li>■ status[7]: Divide-by-zero flag for logarithm functions.</li></ul> |

Table 1-2 Parameter Description

| Parameter | Values       | Description                                                                   |
|-----------|--------------|-------------------------------------------------------------------------------|
| sig_width | 2 to 59 bits | Word length of fraction field of floating-point numbers <i>a</i> and <i>z</i> |
| exp_width | 3 to 31 bits | Width of input and output data buses                                          |

**Table 1-2 Parameter Description (Continued)**

| Parameter       | Values                                      | Description                                                                                                                                                                 |
|-----------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ieee_compliance | 0 or 1                                      | When 1, the generated architecture is capable of dealing with denormals and NaNs                                                                                            |
| extra_prec      | 0 to (59 - <i>sig_width</i> )<br>Default: 0 | Internal extra precision (in bits) used in the computation of the FP output                                                                                                 |
| arch            | 0 to 2<br>Default: 2                        | Implementation selection: <ul style="list-style-type: none"> <li>0 = Area optimized</li> <li>1 = Speed optimized</li> <li>2 = Implementation released in 2007.12</li> </ul> |

**Table 1-3 Synthesis Implementations**

| Implementation Name | Function                                                                                          | License Feature Required |
|---------------------|---------------------------------------------------------------------------------------------------|--------------------------|
| rtl                 | Implementation using the Datapath Generator technology combined with static DesignWare components | DesignWare               |

**Table 1-4 Simulation Model**

| Model                          | Function                             |
|--------------------------------|--------------------------------------|
| DW04.DW_FP_LOG2_CFG_SIM        | Design unit name for VHDL simulation |
| dw/dw04/src/DW_fp_log2_sim.vhd | VHDL simulation model source code    |
| dw/sim_ver/DW_fp_log2.v        | Verilog simulation model source code |

Differently from other FP components, DW\_fp\_log2 does not have rounding mode control, given the properties of algorithms used to compute the logarithms and the goal to deliver the best possible QoR.

The component is able to deliver outputs with a maximum of 2 ulps error for most of the FP range. The special case when the input is in the vicinity of 1.0, and therefore the output approaches 0.0, requires extra hardware just to deal with it. For this reason a new parameter (*extra\_prec*) was introduced.

When the *extra\_pre* parameter is 0, the implementation uses the minimum number of bits required to guarantee a maximum error of 2 ulps for most of the calculated values. When the input value is near 1.0, the output may exhibit larger relative errors.

By using non-zero values for *extra\_prec* the designer gets smaller errors when the input is near 1, but the error bound for other input values continues to be a maximum of 2 ulps. The designer should consider using values of *extra\_prec* in the range [0,*sig\_width*]. When *extra\_prec* = *sig\_width*, the error is bounded to a maximum of 2 ulps for all inputs. Values larger than *sig\_width* do not improve accuracy in any way.

The use of *extra\_prec* affects QoR and the designer must experiment with it to reach a good compromise between accuracy around the input value 1.0 and QoR.

The *arch* parameter controls implementation alternatives for this component. Different values result in different numerical behavior. You should experiment with this parameter to find out which value provides the best QoR for your design constraints and technology. Using *arch* = 0 (area optimized implementation) usually provides the best QoR for most time constraints.

## Alternative Implementation of Floating-point Base-2 Logarithm with DW\_lp\_fp\_multifunc

The floating-point base-2 logarithm operation can also be implemented by DW\_lp\_fp\_multifunc component, which evaluates the value of floating-point base-2 logarithm with a maximum of 2 ulps error bound. There will be a maximum of 2 ulps difference between the value from DW\_lp\_fp\_multifunc and the value from DW\_fp\_log2. Performance and area of the synthesis results are different between the DW\_fp\_log2 and base-2 logarithm implementation of the DW\_lp\_fp\_multifunc, depending on synthesis constraints, library cells and synthesis environments. By comparing performance and area between the base-2 logarithm implementation of DW\_lp\_fp\_multifunc and DW\_fp\_log2 component, the DW\_lp\_fp\_multifunc provides more choices for the better synthesis results. Below is an example of the Verilog description for the floating-point base-2 logarithm of the DW\_lp\_fp\_multifunc. For more detailed information, see the [DW\\_lp\\_fp\\_multifunc](#) datasheet.

```
DW_lp_fp_multifunc #(sig_width, exp_width, ieee_compliance, 32) U1 (
    .a(a),
    .func(16'h0020),
    .rnd(3'h0),
    .z(z),
    .status(status)
);
```

For more information on the floating-point system defined for all the floating-point components in the DesignWare Library, including status flag bits and floating-point formats, refer to the [Datapath Floating-Point Overview](#).

## Simulation

In most cases, the simulation model for DW\_fp\_log2 delivers results that are bit-accurate with the synthesis model. Note the following details.

For Verilog simulations:

- When using VCS:
  - Results are bit-accurate with the synthesis model for all component configurations.
- When using simulators other than VCS:
  - If (sig\_width + extra\_prec + 1) <= 38, results are bit-accurate with the synthesis model
  - If (sig\_width + extra\_prec + 1) > 38, results are not bit-accurate with the synthesis model

This scenario arises inside the DW\_log2 component, which is a submodule of DW\_fp\_log2. For information about the DW\_log2 component, see the [datasheet](#).

For VHDL simulations:

- When (sig\_width + extra\_prec + 1) <= 38, results are bit-accurate with the synthesis model
- When (sig\_width + extra\_prec + 1) > 38, results are not bit-accurate with the synthesis model

This scenario arises inside the DW\_log2 component, which is a submodule of DW\_fp\_log2. For information about the DW\_log2 component, see the [datasheet](#).

## Related Topics

- [Datapath Floating-Point Overview](#)
- [DesignWare Building Block IP User Guide](#)

## HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.dw_foundation_comp.all;

entity DW_fp_log2_inst is
    generic (
        inst_sig_width : POSITIVE := 23;
        inst_exp_width  : POSITIVE := 8;
        inst_ieee_compliance : INTEGER := 0;
        inst_extra_prec  : INTEGER := 0;
        inst_arch        : INTEGER := 2
    );
    port (
        inst_a : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
        z_inst : out std_logic_vector(inst_sig_width+inst_exp_width downto 0);
        status_inst : out std_logic_vector(7 downto 0)
    );
end DW_fp_log2_inst;

architecture inst of DW_fp_log2_inst is
begin

    -- Instance of DW_fp_log2
    U1 : DW_fp_log2
        generic map ( sig_width => inst_sig_width,
                      exp_width  => inst_exp_width,
                      ieee_compliance => inst_ieee_compliance,
                      extra_prec  => inst_extra_prec,
                      arch        => inst_arch)
        port map ( a => inst_a, z => z_inst, status => status_inst );

end inst;

-- pragma translate_off
configuration DW_fp_log2_inst_cfg_inst of DW_fp_log2_inst is
    for inst
    end for; -- inst
end DW_fp_log2_inst_cfg_inst;
-- pragma translate_on
```

## HDL Usage Through Component Instantiation - Verilog

```
module DW_fp_log2_inst( inst_a, z_inst, status_inst );

parameter sig_width = 10;
parameter exp_width = 5;
parameter ieee_compliance = 0;
parameter extra_prec = 0;
parameter arch = 2;

input [sig_width+exp_width : 0] inst_a;
output [sig_width+exp_width : 0] z_inst;
output [7 : 0] status_inst;

    // Instance of DW_fp_log2
    DW_fp_log2 #(sig_width, exp_width, ieee_compliance, extra_prec, arch)
        U1 ( .a(inst_a), .z(z_inst), .status(status_inst) );

endmodule
```

## Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

| Date           | Release       | Updates                                                                                                                                                                                                                                                                                                     |
|----------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| September 2024 | DWBB_202409.0 | <ul style="list-style-type: none"> <li>Updated date and version; no technical changes</li> </ul>                                                                                                                                                                                                            |
| December 2023  | DWBB_202312.0 | <ul style="list-style-type: none"> <li>Updated date and version; no technical changes</li> </ul>                                                                                                                                                                                                            |
| October 2022   | DWBB_202203.5 | <ul style="list-style-type: none"> <li>Clarified error bounding in ulps on <a href="#">page 2</a> and <a href="#">page 3</a></li> </ul>                                                                                                                                                                     |
| September 2022 | DWBB_202203.4 | <ul style="list-style-type: none"> <li>Added “<a href="#">Simulation</a>” on <a href="#">page 3</a> to explain how results from the simulation model match the synthesis model</li> </ul>                                                                                                                   |
| January 2020   | DWBB_201912.1 | <ul style="list-style-type: none"> <li>Corrected port names for DW_lp_fp_multifunc in “<a href="#">Alternative Implementation of Floating-point Base-2 Logarithm with DW_lp_fp_multifunc</a>” on <a href="#">page 3</a></li> </ul>                                                                          |
| December 2019  | DWBB_201912.0 | <ul style="list-style-type: none"> <li>Adjusted the upper value bound of <i>extra_prec</i> in <a href="#">Table 1-2</a> on <a href="#">page 1</a></li> </ul>                                                                                                                                                |
| July 2019      | DWBB_201903.3 | <ul style="list-style-type: none"> <li>Removed reference to minPower library in “<a href="#">Alternative Implementation of Floating-point Base-2 Logarithm with DW_lp_fp_multifunc</a>” on <a href="#">page 3</a></li> <li>Added this Revision History table and the document links on this page</li> </ul> |

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